task 6

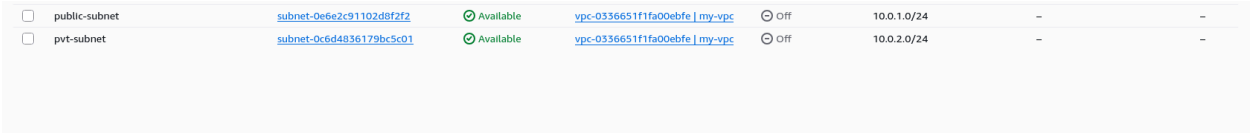
Create Gateway VPC Endpoint for S3 Remove NAT Gateway Ensure private EC2 can: * Access S3 ∗ Cannot access internet Validate using CLI

Steps:

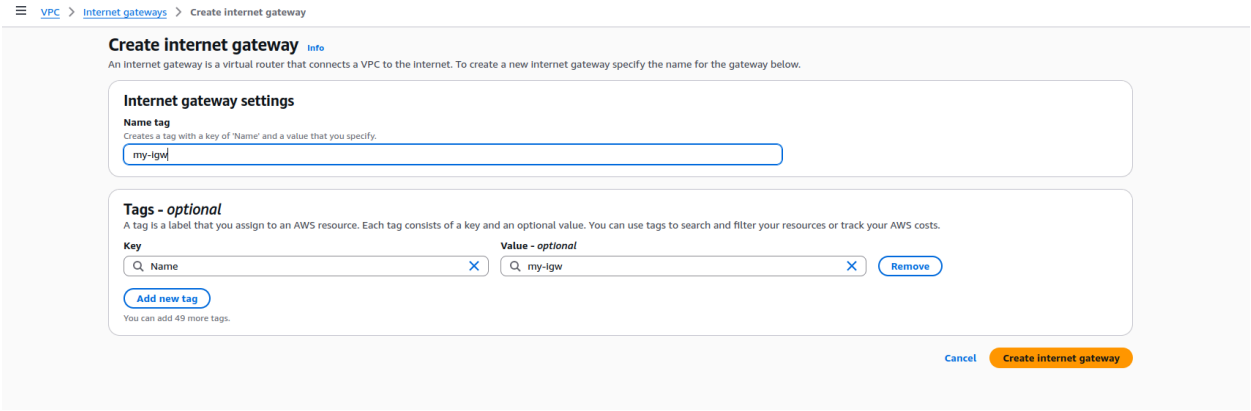
1. Create VPC
   - CIDR: `10.0.0.0/16`
2. Create Subnets
   - Public Subnet: `10.0.1.0/24`
   - Private Subnet: `10.0.2.0/24`
3. Create Internet Gateway
   - Attach to VPC
4. Create Route Tables
   - Public Route Table → add route `0.0.0.0/0 → IGW` → associate with public subnet
   - Private Route Table → NO internet route → associate with private subnet
5. Launch EC2 Instance
   - Place in **private subnet**
   - Disable public IP
6. Ensure NAT Gateway
   - Do NOT create OR delete if exists
7. Create S3 Bucket
8. Create VPC Endpoint

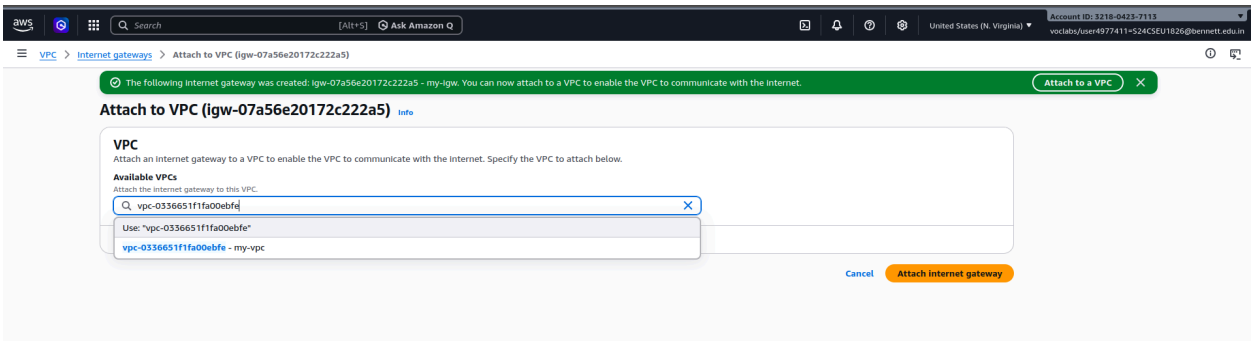
- Type: Gateway
- Service: S3
- Attach to VPC
- Select **private route table**

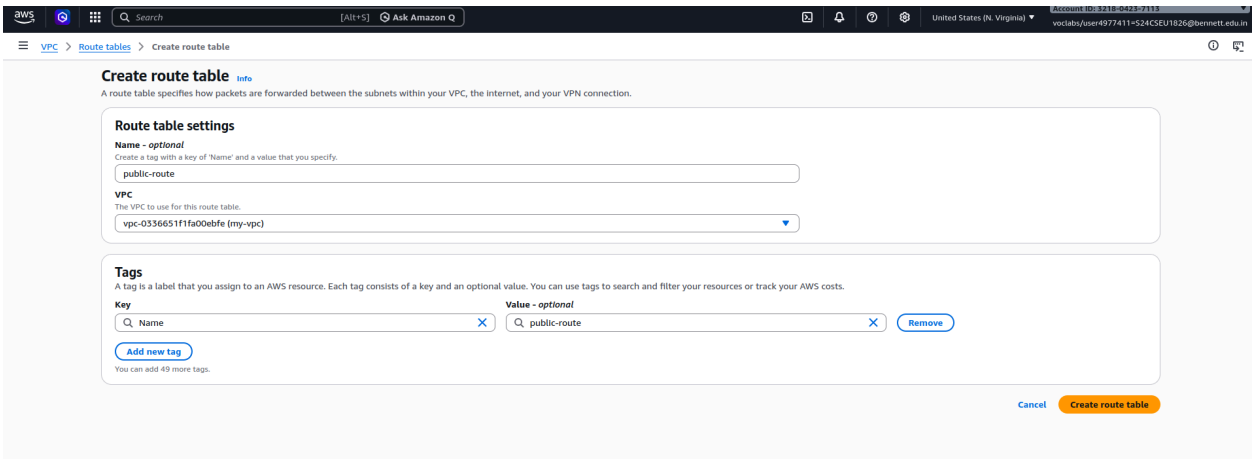
9. Verify
    - Private route table has
    - No route
    - EC2 has no public IP

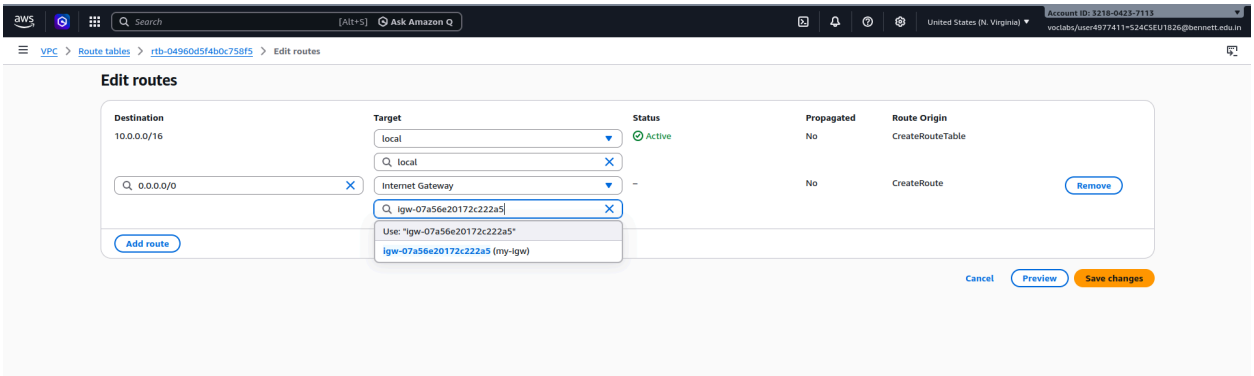


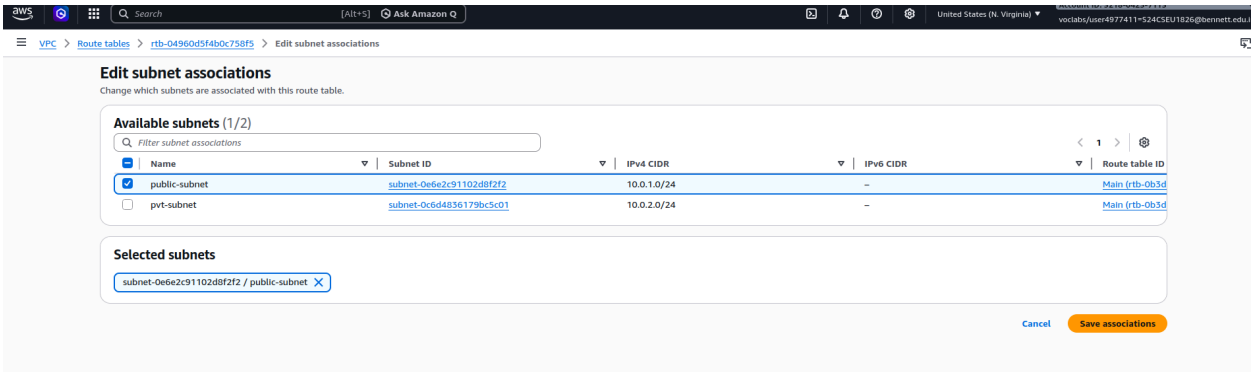
public-subnet	subnet-0e6e2c91102d8f2f2	Available	vpc-0336651f1fa00ebfe | my-vpc	Off	10.0.1.0/24	–	–
pvt-subnet	subnet-0c6d4836179bc5c01	Available	vpc-0336651f1fa00ebfe | my-vpc	Off	10.0.2.0/24	–	–

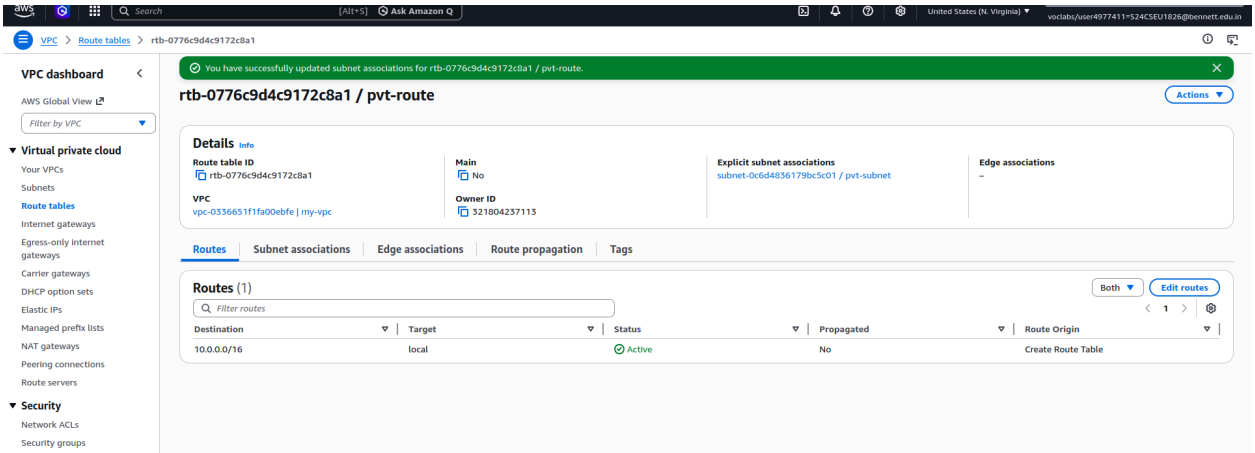


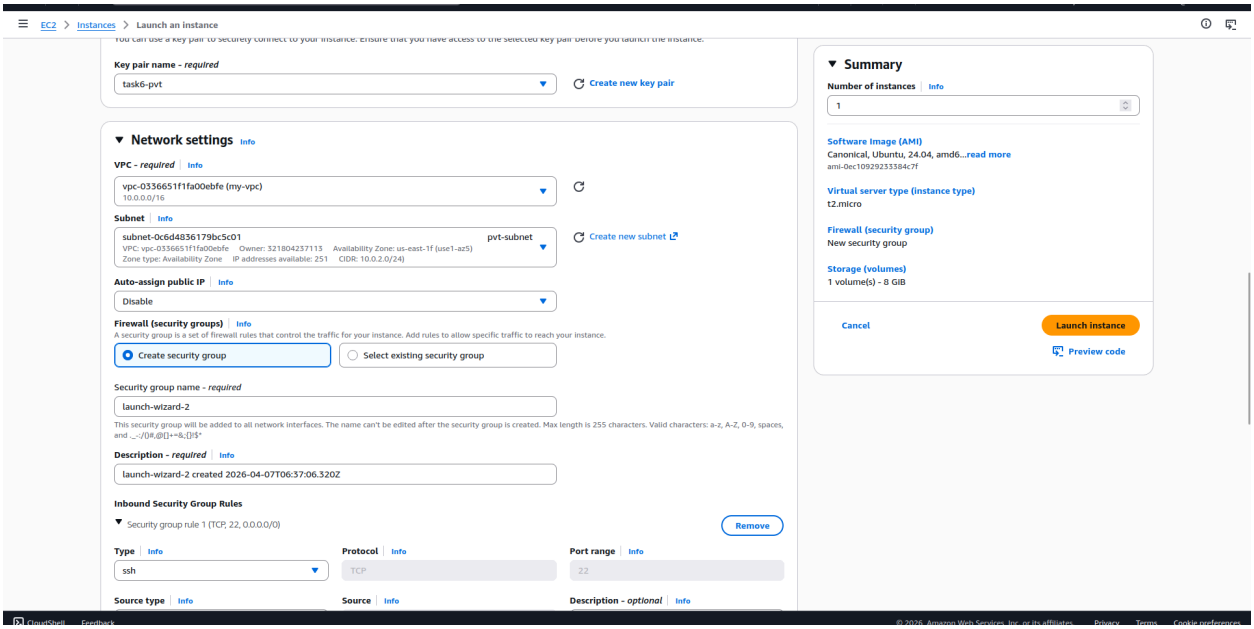

Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 3218-0423-7113
voclabs/user4977411=S24CSEU1826@bennett.edu.in
VPC > Internet gateways > Attach to VPC (igw-07a56e20172c222a5)
The following Internet gateway was created: igw-07a56e20172c222a5 - my-igw. You can now attach to a VPC to enable the VPC to communicate with the Internet.
Attach to a VPC
Attach to VPC (igw-07a56e20172c222a5) Info
VPC
Attach an Internet gateway to a VPC to enable the VPC to communicate with the Internet. Specify the VPC to attach below.
Available VPCs
Attach the Internet gateway to this VPC.
vpc-0336651f1fa00ebfe
Use: "vpc-0336651f1fa00ebfe"
vpc-0336651f1fa00ebfe - my-vpc
Cancel
Attach internet gateway

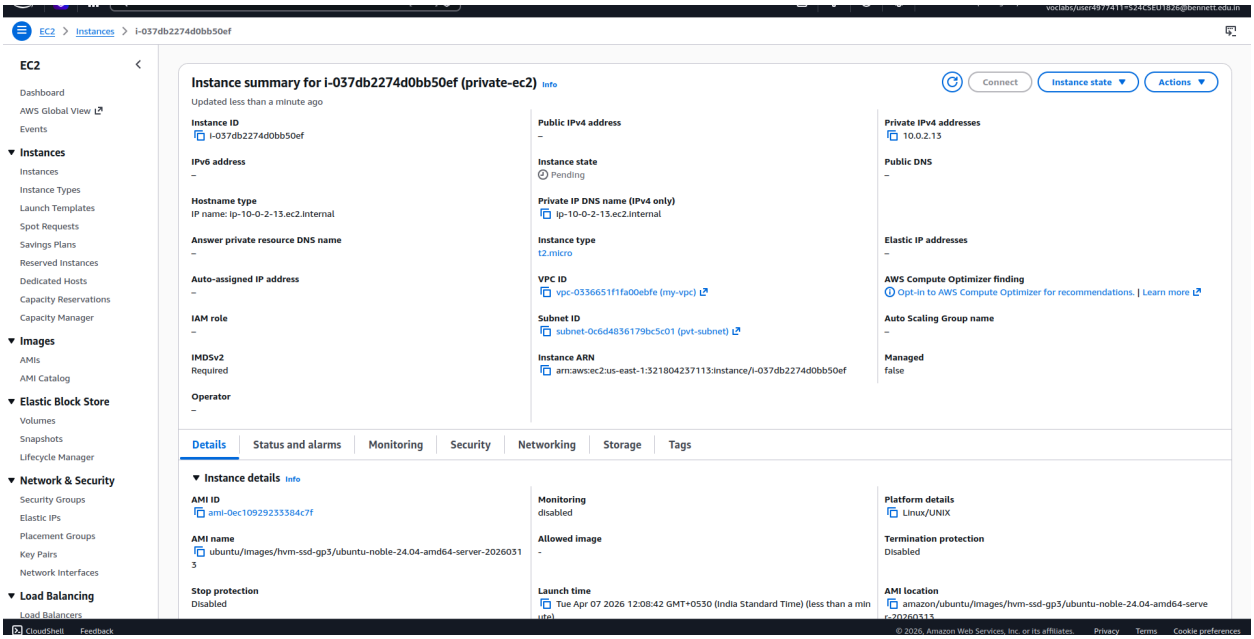

Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 3218-0423-7113
voclabs/user4977411=S24CSEU1826@bennett.edu.in
VPC > Route tables > Create route table
Create route table Info
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
public-route
VPC
The VPC to use for this route table.
vpc-0336651f1fa00ebfe (my-vpc)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
public-route
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table


Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 3218-0423-7113
voclabs/user4977411=S24CSEU1826@bennett.edu.in
VPC > Route tables > rtb-04960d5f4b0c758f5 > Edit routes
Edit routes
Destination
Target
Status
Propagated
Route Origin
10.0.0.0/16
local
Active
No
CreateRouteTable
local
0.0.0.0/0
Internet Gateway
-
No
CreateRoute
Remove
igw-07a56e20172c222a5
Use: "igw-07a56e20172c222a5"
igw-07a56e20172c222a5 (my-igw)
Add route
Cancel
Preview
Save changes


VPC > Route tables > rtb-04960d5f4b0c758f5 > Edit subnet associations
Edit subnet associations
Change which subnets are associated with this route table.
Available subnets (1/2)
Filter subnet associations
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
Route table ID
public-subnet
subnet-0e6e2c91102d8f2f2
10.0.1.0/24
Main (rtb-0b3d
pvt-subnet
subnet-0c6d4836179bc5c01
10.0.2.0/24
Main (rtb-0b3d
Selected subnets
subnet-0e6e2c91102d8f2f2 / public-subnet
Cancel
Save associations


VPC > Route tables > rtb-0776c9d4c9172c8a1
You have successfully updated subnet associations for rtb-0776c9d4c9172c8a1 / pvt-route.
rtb-0776c9d4c9172c8a1 / pvt-route
Actions
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
Details Info
Route table ID
rtb-0776c9d4c9172c8a1
Main
No
Explicit subnet associations
subnet-0c6d4836179bc5c01 / pvt-subnet
Edge associations
VPC
vpc-0336651f1fa00ebfe | my-vpc
Owner ID
321804237113
Routes
Subnet associations
Edge associations
Route propagation
Tags
Routes (1)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
10.0.0.0/16
local
Active
No
Create Route Table


EC2 > Instances > Launch an instance
Key pair name - required
task6-pvt
Create new key pair
Network settings Info
VPC - required Info
vpc-0336651f1fa00ebfe (my-vpc)
10.0.0.0/16
Subnet Info
subnet-0c6d4836179bc5c01
pvt-subnet
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
launch-wizard-2
Description - required Info
launch-wizard-2 created 2026-04-07T06:37:06.320Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 0.0.0.0/0)
Remove
Type Info
ssh
Protocol Info
TCP
Port range Info
22
Source type Info
Source Info
Description - optional Info
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0ec10929233384c7f
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code


EC2 > Instances > i-037db2274d0bb50ef
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Instance summary for i-037db2274d0bb50ef (private-ec2) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-037db2274d0bb50ef
IPv6 address
–
Hostname type
IP name: ip-10-0-2-13.ec2.internal
Answer private resource DNS name
–
Auto-assigned IP address
–
IAM role
–
IMDSv2
Required
Operator
–
Public IPv4 address
–
Instance state
Pending
Private IP DNS name (IPv4 only)
ip-10-0-2-13.ec2.internal
Instance type
t2.micro
VPC ID
vpc-0336651f1fa00ebfe (my-vpc)
Subnet ID
subnet-0c6d4836179bc5c01 (pvt-subnet)
Instance ARN
arn:aws:ec2:us-east-1:321804237113:instance/i-037db2274d0bb50ef
Private IPv4 addresses
10.0.2.13
Public DNS
–
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
Auto Scaling Group name
–
Managed
false
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance details Info
AMI ID
ami-0ec10929233384c7f
AMI name
ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20260313
Stop protection
Disabled
Monitoring
disabled
Allowed image
-
Launch time
Tue Apr 07 2026 12:08:42 GMT+0530 (India Standard Time) (less than a minute)
Platform details
Linux/UNIX
Termination protection
Disabled
AMI location
amazon/ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20260313
CloudShell
Feedback
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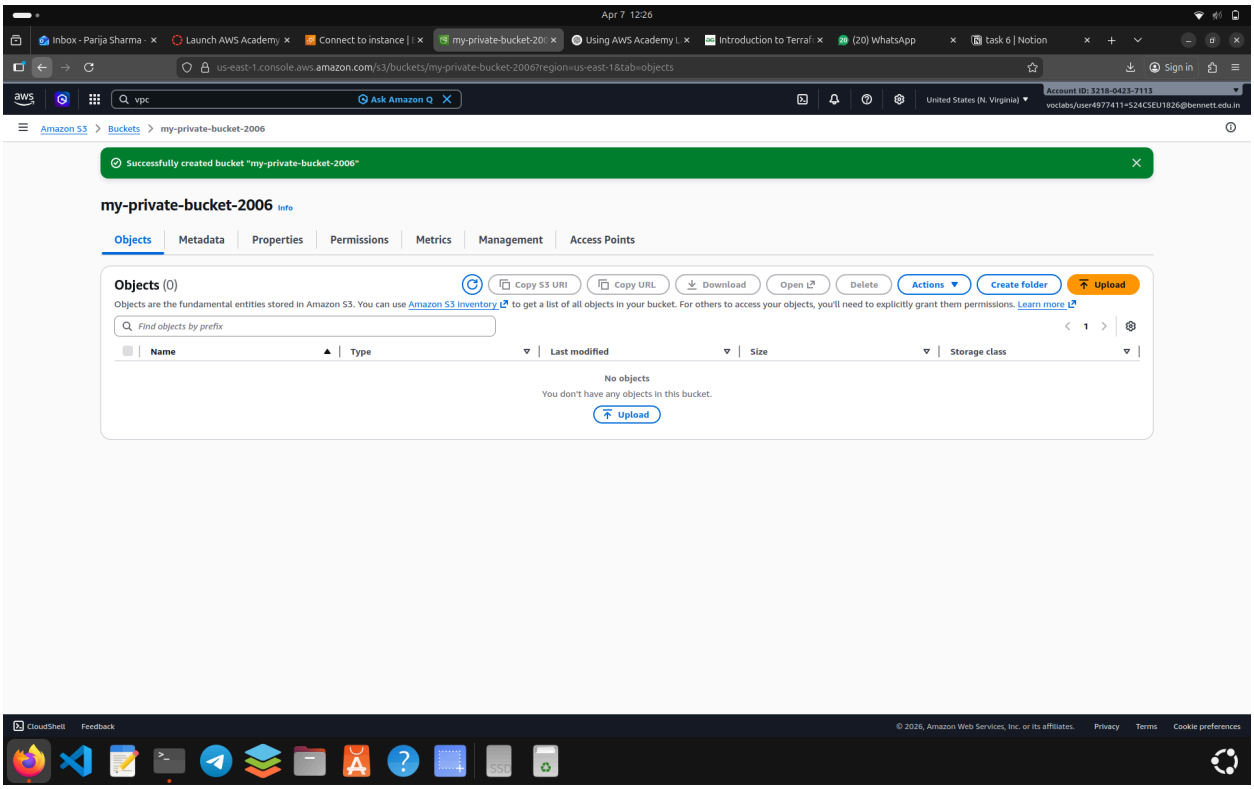
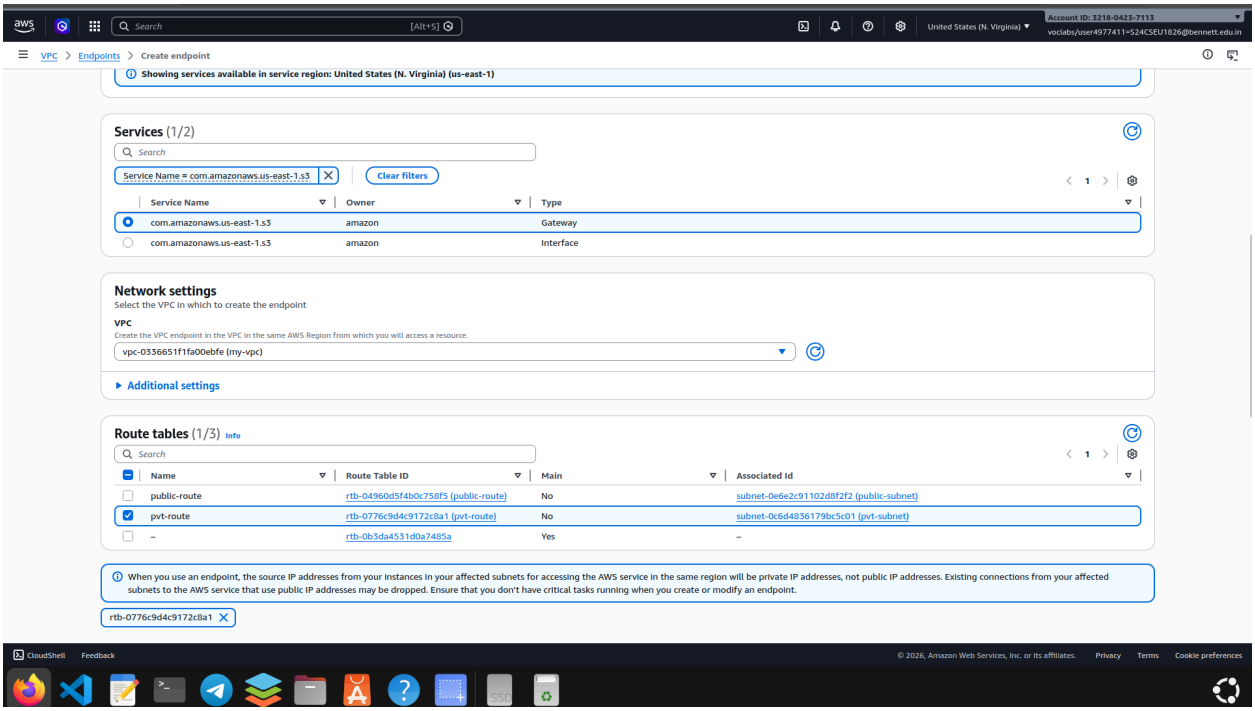
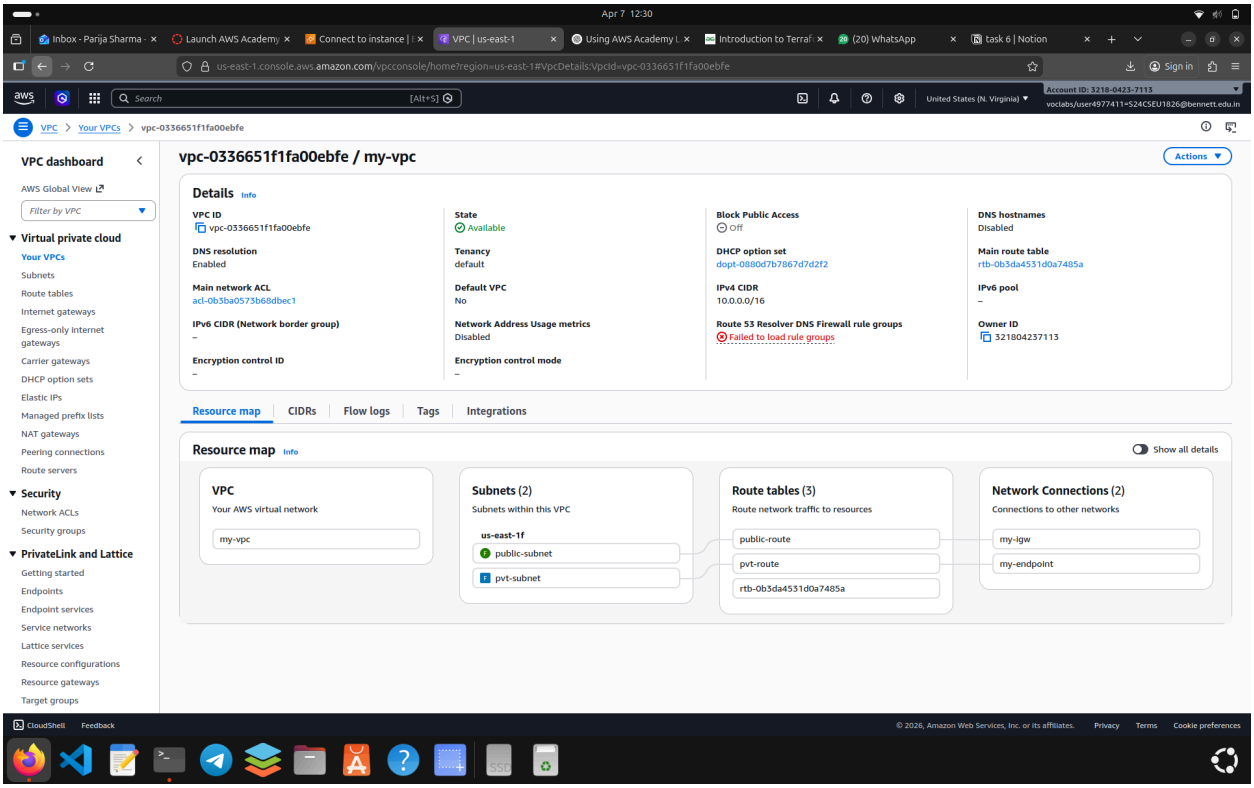
Privacy
Terms
Cookie preferences


Successfully created bucket "my-private-bucket-2006"
my-private-bucket-2006
Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 Inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload


Showing services available in service region: United States (N. Virginia) (us-east-1)
Services (1/2)
Service Name = com.amazonaws.us-east-1.s3
Clear filters
Service Name
Owner
Type
com.amazonaws.us-east-1.s3 amazon Gateway
com.amazonaws.us-east-1.s3 amazon Interface
Network settings
Select the VPC in which to create the endpoint
VPC
Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.
vpc-0336651f1fa00ebfe (my-vpc)
Additional settings
Route tables (1/3)
Name
Route Table ID
Main
Associated Id
public-route rtb-04960d5f4b0c758f5 (public-route) No subnet-0e6e2c91102d8f2f2 (public-subnet)
pvt-route rtb-0776c9d4c9172c8a1 (pvt-route) No subnet-0c6d4836179bc5c01 (pvt-subnet)
– rtb-0b3da4531d0a7485a Yes –
When you use an endpoint, the source IP addresses from your instances in your affected subnets for accessing the AWS service in the same region will be private IP addresses, not public IP addresses. Existing connections from your affected subnets to the AWS service that use public IP addresses may be dropped. Ensure that you don't have critical tasks running when you create or modify an endpoint.
rtb-0776c9d4c9172c8a1


Apr 7 12:30
Inbox - Parija Sharma
Launch AWS Academy
Connect to instance
VPC | us-east-1
Using AWS Academy L
Introduction to Terraf
(20) WhatsApp
task 6 | Notion
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#VpcDetails:VpcId=vpc-0336651f1fa00ebfe
Sign in
Search
[Alt+S]
United States (N. Virginia)
Account ID: 3218-0423-7113
voclabs/user4977411=S24C5EU1826@bennett.edu.in
VPC > Your VPCs > vpc-0336651f1fa00ebfe
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services
Resource configurations
Resource gateways
Target groups
vpc-0336651f1fa00ebfe / my-vpc
Actions
Details Info
VPC ID
vpc-0336651f1fa00ebfe
State
Available
Block Public Access
Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-0880d7b7867d7d2f2
Main route table
rtb-0b3da4531d0a7485a
Main network ACL
acl-0b3ba0573b68dbec1
Default VPC
No
IPv4 CIDR
10.0.0.0/16
IPv6 pool
–
IPv6 CIDR (Network border group)
–
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
Failed to load rule groups
Owner ID
321804237113
Encryption control ID
–
Encryption control mode
–
Resource map
CIDRs
Flow logs
Tags
Integrations
Resource map Info
Show all details
VPC
Your AWS virtual network
my-vpc
Subnets (2)
Subnets within this VPC
us-east-1f
public-subnet
pvt-subnet
Route tables (3)
Route network traffic to resources
public-route
pvt-route
rtb-0b3da4531d0a7485a
Network Connections (2)
Connections to other networks
my-igw
my-endpoint
CloudShell
Feedback
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